

Zero Partial-Period Correlation Zone Sequence Sets for MIMO Channel Estimation

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Abstract—In this paper we propose zero partial-period correlation zone (ZPCZ) sequences which may be considered as a generalized version of conventional zero-correlation zone (ZCZ) sequences. We present their basic properties and apply them to channel estimation. When they are employed as training sequences for multiple-input multiple-output (MIMO) channel estimation, the mean-square error (MSE) of the minimum mean-square error (MMSE) estimator can be minimized.

Index Terms—Channel estimation, partial-period correlation, zero-correlation zone (ZCZ) sequences.

I. INTRODUCTION

Pseudorandom sequences with good correlation properties have many applications to communication systems. In order to increase the performance of these systems, it is required for an employed family of pseudorandom sequences to have large size and low correlation. There are three types of correlation: periodic correlation, aperiodic correlation, and partial-period correlation. Partial-period correlation for some sequences was studied in [1]–[3]. It is well-known that the expectations of the partial-period cross-correlation values of periodic sequences are proportional to their periodic cross-correlation values [3]. Periodic sequences with low partial-period correlation are desirable in applications such as spread-spectrum communications [4] and channel estimations.

For wireless communication systems, it is very important to estimate the channels accurately for reliable transmission of data. In multiple-input multiple-output (MIMO) systems, efficient channel estimation is much more challenging because of the existence of the interference among multiple antennas. One of the most widely used schemes for channel estimation is to employ known training sequences [5]. By using training sequences which are easily distinguishable, the channel estimator at the receiver is able to make an estimate of the channel impulse response (CIR) [6].

There are two common training-based channel estimation schemes for MIMO systems: one is the time-division multiplexing (TDM) scheme, and the other is the implicit estimation scheme. In the training-based TDM scheme, training sequences are assigned in a limited number of time slots. In [7], Yang and Wu derived the condition for the TDM training sequences in a MIMO channel to make each link independent. Fragouli, *et al.* [8] also presented optimal training sequences

in a similar approach and called them L-perfect training sequences. On the other hand, training sequences in implicit MIMO estimation schemes are combined with information in every time slot. For these schemes, zero-correlation zone (ZCZ) sequences [9] based on full-period correlation may be used as training sequences.

In this paper, we propose zero partial-period correlation zone (ZPCZ) sequences as TDM training sequences for MIMO channel estimation. They have zero partial-period correlation within a given zone. When they are applied as training sequences, the mean-square error (MSE) of the minimum mean-square error (MMSE) estimator can be minimized.

The outline of the paper is as follows. Section II presents ZPCZ sequences and their properties and gives a construction method. In Section III, we propose an application of ZPCZ sequences to MIMO channel estimation. Finally, we give some concluding remarks in Section IV.

II. ZERO PARTIAL-PERIOD CORRELATION ZONE SEQUENCES

A. Partial-Period Correlation

Let $\{a(n)\}$ and $\{b(n)\}$ be two complex-valued sequences of period N . The (full-period) periodic cross-correlation $\theta_{a,b}(\tau)$ between $\{a(n)\}$ and $\{b(n)\}$ is defined as

$$\theta_{a,b}(\tau) = \sum_{n=0}^{N-1} a(n)b^*(n+\tau)$$

for an integer $0 \leq \tau \leq N-1$, where $*$ denotes the complex conjugation and all the operations among the indices are computed modulo N . The aperiodic cross-correlation $A_{a,b}(\tau)$ between $\{a(n)\}$ and $\{b(n)\}$ is defined by

$$A_{a,b}(\tau) = \sum_{n=\max\{0, -\tau\}}^{\min\{N-1, N-1-\tau\}} a(n)b^*(n+\tau)$$

for an integer $0 \leq \tau \leq N-1$.

As another measure, the partial-period correlation may be employed.

Definition 1 ([3]). For an integer $0 \leq k < N$ and an integer $0 < B \leq N$, the partial-period correlation $R_{a,b}(\tau, k, B)$ between $\{a(n)\}$ and $\{b(n)\}$ is defined as

$$R_{a,b}(\tau, k, B) = \sum_{n=k}^{B+k-1} a(n)b^*(n+\tau),$$

where all the operations among the indices are computed modulo N .

When $a(n) = b(n)$ for any integer n , we call $R_{a,b}(\tau, k, B)$ the partial-period autocorrelation of $\{a(n)\}$, denoted by $R_a(\tau, k, B)$. Note that the partial-period correlation is not parameterized only by the shift τ , but also by the initial position k and the length B of the correlation window. One can easily check that k does not affect the correlation values any more, if $B = N$. With additional parameters k and B , the partial-period correlation is related to the periodic and aperiodic correlations via

$$\begin{aligned} \theta_{a,b}(\tau) &= R_{a,b}(\tau, 0, N), \\ A_{a,b}(\tau) &= R_{a,b}(\tau, \max\{0, -\tau\}, N - |\tau|). \end{aligned}$$

Definition 2. Let $\mathcal{S} = \{\{s_i(n)\} | 1 \leq i \leq M\}$ be a set of M sequences of period N and Z a positive integer. If $\mathcal{S}$ satisfies

$$R_{s_i, s_j}(\tau, k, B) = \begin{cases} B, & \text{for } \tau = 0 \text{ when } i = j \\ 0, & \text{for } 0 < |\tau| < Z \text{ when } i = j \\ 0, & \text{for } 0 \leq |\tau| < Z \text{ when } i \neq j, \end{cases}$$

for any k such that $\max\{-\tau, 0\} \leq k \leq \min\{Z-1-\tau, Z-1\}$, then it is called a zero partial-period correlation zone (ZPCZ) sequence set, denoted by an (N, M, Z, B) -ZPCZ sequence set.

Clearly, if $B = N$, any (N, M, Z, B) -ZPCZ sequence set is reduced to an (N, M, Z) -ZCZ sequence set defined in [10]. Therefore, any conventional ZCZ sequence set may be considered as a special case of ZPCZ sequence sets, where $B = N$.

B. Basic Properties

In this subsection, we will give some basic properties of partial-period correlation and ZPCZ sequence sets. First, the (full-period) periodic cross-correlation $\theta_{a,b}(\tau)$ is always expressed as the sum of three partial-period cross-correlation values as follows:

$$\begin{aligned} \theta_{a,b}(\tau) &= R_{a,b}(\tau, 0, k_0) + R_{a,b}(\tau, k_0, B_0) \\ &\quad + R_{a,b}(\tau, k_0 + B_0, N - B_0 - k_0) \end{aligned}$$

for some integers k_0 and B_0 such that $k_0 \geq 1$ and $k_0 + B_0 \leq N$, where all the operations among the indices are computed modulo N . Furthermore, $R_{a,b}(\tau)$ has the following conjugate symmetry-after-shift property

$$R_{a,b}(\tau, k, B) = R_{b,a}^*(-\tau, k + \tau, B).$$

Proposition 3. Let $\mathcal{S} = \{\{s_i(n)\} | 1 \leq i \leq M\}$ be an (N, M, Z, B) -ZPCZ sequence set over the set of complex numbers with magnitude 1. Then the followings hold:

- (a) The length B of the correlation window must be even.
- (b) B is lower-bounded by the product of M and Z , i.e.,

$$B \geq MZ.$$

Note that (b) in Proposition 3 can be viewed as a special case of the results in [11].

C. A Simple Construction of ZPCZ Sequence Sets

A $(P + Z - 1, M, Z, P)$ -ZPCZ sequence set may be easily obtained from a conventional (P, M, Z) -ZCZ sequence set. Let $\mathcal{A} = \{\{a_i(n)\} | 1 \leq i \leq M\}$ be a (P, M, Z) -ZCZ sequence set. That is, the sequences belonging to $\mathcal{A}$ satisfy

$$\begin{aligned} \theta_{a_i, a_j}(\tau) &= \sum_{n=0}^{P-1} a_i(n) a_j^*(n + \tau \bmod P) \\ &= \begin{cases} P, & \text{if } \tau = 0 \text{ and } i = j \\ 0, & \text{for } 0 < |\tau| < Z \text{ when } i = j \\ 0, & \text{for } 0 \leq |\tau| < Z \text{ when } i \neq j. \end{cases} \end{aligned}$$

Define $\mathcal{S} = \{\{s_i(n)\} | 1 \leq i \leq M\}$ as the set of sequences of period $N = P + Z - 1$, where $s_i(n)$ is obtained by repeating the first $Z - 1$ components of $a_i(n)$ as follows:

$$s_i(n) = a_i(n \bmod P)$$

for $0 \leq n \leq P + Z - 2$.

Theorem 4. $\mathcal{S}$ is a $(P + Z - 1, M, Z, P)$ -ZPCZ set.

Proof. The partial-period cross-correlation between $\{s_i(n)\}$ and $\{s_j(n)\}$ in $\mathcal{S}$ with $B = P$ is computed as follows:

$$\begin{aligned} R_{s_i, s_j}(k, \tau, P) &= \sum_{n=k}^{P+k-1} s_i(n) s_j^*(n + \tau) \\ &= \sum_{n=0}^{P-1} s_i(n + k) s_j^*(n + k + \tau) \\ &= \sum_{n=0}^{P-1} a_i(n + k \bmod P) a_j^*(n + k + \tau \bmod P) \\ &= \sum_{n=0}^{P-1} a_i(n) a_j^*(n + \tau \bmod P) \\ &= \theta_{a_i, a_j}(\tau) \end{aligned}$$

Therefore, the set $\mathcal{S}$ is a $(P + Z - 1, M, Z, P)$ -ZPCZ sequence set. $\square$

III. APPLICATION TO MIMO CHANNEL ESTIMATION

As an application of ZPCZ sequences, we present a training-based TDM scheme for MIMO channel estimation. Since MIMO systems require more accurate demodulation than SISO systems, an accurate channel estimation is essential [12]. In this section, we will give a MIMO system model and show that ZPCZ sequences are optimal training sequences in the sense that the MSE of the MMSE channel estimator is minimized.

A. MIMO System Model

Consider a block transmission MIMO system with N_t transmit and N_r receive antennas over a frequency-selective channel. Let $\{x_i(n)\}$ be a block of length N transmitted from the i th transmit antenna, and let the sequence $\{h_{ji}(n)|n = 0, 1, \dots, L-1\}$ denote the CIR between the i th transmitter and the j th receiver, where L is the channel length. We assume that the channel coefficients remain constant during the transmission of a block and vary independently from block to block [8]. It is also assumed that L is known to the receiver. Let $\{y_j(n)\}$ and $\{w_j(n)\}$ be the received sequence of length N and the corresponding additive white Gaussian noise (AWGN) at j th receiver, respectively. Assuming perfect synchronization, we represent each sequence in vector notation for convenience as follows:

$$\begin{aligned} \mathbf{x}_i &= [x_i(0), \dots, x_i(P-1), x_i(P), \dots, x_i(N-1)]^T, \\ \mathbf{h}_{ji} &= [h_{ji}(0), \dots, h_{ji}(L-1)]^T, \\ \mathbf{w}_j &= [w_j(0), \dots, w_j(P-1), w_j(P), \dots, w_j(N-1)]^T, \\ \mathbf{y}_j &= [y_j(0), \dots, y_j(P-1), y_j(P), \dots, y_j(N-1)]^T \end{aligned}$$

where $P < N$.

In the TDM scheme where each transmit antenna has a training sequence in the head part of each transmitted data block, we assume that $\mathcal{C} = \{\{c_i(n)\}|1 \leq i \leq N_t\}$ is a set of N_t training sequences of period P assigned to N_t transmit antennas. Then the input sequence $\{x_i(n)\}$ consists of P training symbols and $N - P$ information symbols, that is, $\mathbf{x}_i$ can be represented as

$$\mathbf{x}_i = [c_i(0), \dots, c_i(P-1), x_i(P), \dots, x_i(N-1)]^T$$

Throughout this paper, we assume that $L \leq P$.

B. MMSE Channel Estimation

Let $\tilde{\mathbf{y}}_j$ and $\tilde{\mathbf{w}}_j$ be the truncated vectors of $\mathbf{y}_j$ and $\mathbf{w}_j$, respectively, of length $P - L + 1$ as follows:

$$\begin{aligned} \tilde{\mathbf{y}}_j &= [y_j(L-1), \dots, y_j(P-1)]^T, \\ \tilde{\mathbf{w}}_j &= [w_j(L-1), \dots, w_j(P-1)]^T \end{aligned}$$

Then we have

$$y_j(n) = \sum_{i=1}^{N_t} \sum_{k=0}^{L-1} h_{ji}(k) c_i(n-k) + w_j(n), \quad n = L-1, \dots, P-1.$$

Using the knowledge of $\{c_i(n)\}$, it is easily shown that

$$\tilde{\mathbf{y}}_j = \mathbf{C} \mathbf{H}_j + \tilde{\mathbf{w}}_j$$

where

$$\mathbf{H}_j = [h_{j1}(0), \dots, h_{j1}(L-1), \dots, h_{jN_t}(0), \dots, h_{jN_t}(L-1)]^T$$

Here, note that $\mathbf{C}$ is a $(P - L + 1) \times LN_t$ training matrix consisting of N_t submatrices, given by

$$\mathbf{C} = [\mathbf{C}_1, \dots, \mathbf{C}_{N_t}],$$

where

$$\mathbf{C}_i = \begin{bmatrix} c_i(L-1) & c_i(L-2) & \dots & c_i(0) \\ c_i(L) & c_i(L-1) & \dots & c_i(1) \\ \vdots & \vdots & \ddots & \vdots \\ c_i(P-1) & c_i(P-2) & \dots & c_i(P-L) \end{bmatrix}. \quad (1)$$

If all the LN_t columns of $\mathbf{C}$ are lineally independent, the estimate $\hat{\mathbf{H}}_j$ of $\mathbf{H}_j$ can be obtained by applying the least-square approximation (LSA):

$$\begin{aligned} \hat{\mathbf{H}}_j &= (\mathbf{C}^H \mathbf{C})^{-1} \mathbf{C}^H \tilde{\mathbf{y}}_j \\ &= \mathbf{H}_j + (\mathbf{C}^H \mathbf{C})^{-1} \mathbf{C}^H \tilde{\mathbf{w}}_j \end{aligned} \quad (2)$$

where superscript H is the conjugate transpose. The mean-square error (MSE) of $\hat{\mathbf{H}}_j$ is computed by

$$\text{MSE} = E[(\hat{\mathbf{H}}_j - \mathbf{H}_j)^H (\hat{\mathbf{H}}_j - \mathbf{H}_j)].$$

It is well-known in [7], [8] that the MSE is minimized if and only if

$$\mathbf{C}^H \mathbf{C} = (P - L + 1) \mathbf{I}_{LN_t}, \quad (3)$$

and the corresponding minimum MSE is given by

$$\text{MSE}_{\min} = \frac{LN_t \sigma^2}{P - L + 1}$$

where σ^2 is the variance of AWGN. It follows from (2) that each CIR is estimated independently of other CIRs when (3) is satisfied.

C. ZPCZ Sequences as Training Sequences for MIMO channels

By a simple observation, it is easily checked that the condition in (3) is equivalent to

$$\mathbf{C}_i^H \mathbf{C}_j = \begin{cases} \mathbf{0}_L, & \text{if } i \neq j \\ (P - L + 1) \mathbf{I}_L, & \text{if } i = j \end{cases}$$

for $1 \leq i, j \leq N_t$, where $\mathbf{C}_i$ is of the form in (1). This condition can be achieved by ZPCZ sequences.

Theorem 5. Let $\mathcal{C} = \{\{c_i(n)\}|1 \leq i \leq M\}$ be an (N, M, Z, B) -ZPCZ sequence set, where $B = N - Z + 1$. If the channel length L is less than or equal to the zone size Z , i.e., $L \leq Z$, then $\mathcal{C}$ is a training sequence set achieving the minimum MSE in MIMO channel estimation.

$$\mathbf{C}_i^H \mathbf{C}_j = \begin{bmatrix} R_{c_j, c_i}(0, L-1, B) & R_{c_j, c_i}(1, L-2, B) & \cdots & R_{c_j, c_i}(L-1, 0, B) \\ R_{c_j, c_i}(-1, L-1, B) & R_{c_j, c_i}(0, L-2, B) & \cdots & R_{c_j, c_i}(L-2, 0, B) \\ \vdots & \vdots & \vdots & \vdots \\ R_{c_j, c_i}(-(L-1), L-1, B) & R_{c_j, c_i}(-(L-2), L-2, B) & \cdots & R_{c_j, c_i}(0, 0, B) \end{bmatrix}. \quad (4)$$

Proof. Let $P = N$ and $N_t = M$ in (1). Then the training submatrix $\mathbf{C}_i$ corresponding to $\{c_i(n)\}$ is given by

$$\mathbf{C}_i = \begin{bmatrix} c_i(L-1) & c_i(L-2) & \cdots & c_i(0) \\ c_i(L) & c_i(L-1) & \cdots & c_i(1) \\ \vdots & \vdots & \vdots & \vdots \\ c_i(B+L-2) & c_i(B+L-3) & \cdots & c_i(B-1) \end{bmatrix}.$$

The inner product between any column of $\mathbf{C}_i$ and any column of $\mathbf{C}_j$ is equal to $R_{c_j, c_i}(\tau, k, B)$ for some integers τ and k . Thus, $\mathbf{C}_i^H \mathbf{C}_j$ is given as in (4). From the definition of ZPCZ sequences, we get

$$\mathbf{C}_i^H \mathbf{C}_j = \begin{cases} \mathbf{0}_L, & \text{if } i \neq j \\ B\mathbf{I}_L, & \text{if } i = j, \end{cases}$$

when $L \leq Z$. Therefore, $\mathcal{C}$ can be used as training sequences for MIMO channel estimation with the minimum MSE. $\square$

IV. CONCLUDING REMARKS

Motivated by the training-based TDM scheme for MIMO channel estimation, we proposed the concept of ZPCZ sequences. It was shown that the minimum MSE can be achieved by applying the proposed sequences as training sequences to the TDM scheme. Our future work is to combine the proposed channel estimation method with various coded systems. Finding more novel constructions for efficient ZPCZ sequences may also be an interesting problem.

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